Mark schemes

Q1.

(a) $y = \ln x$

penalise + c once on 1 (a) or 2 (a)

$$\frac{dy}{dx} = \frac{1}{x}$$

B1

1

(b) $y = (x + 1) \ln x$

$$\frac{dy}{dx} = (x+1) \times \frac{1}{x} + \ln x$$

product rule

M1
A1

2

(c) $y = (x + 1)\ln x$

$$\frac{dy}{dx} = \frac{1}{x} + 1 + \ln x$$

$x = 1$: $\frac{dy}{dx} = 1 + 1 = 2$

substitute x = 1 into their $\frac{dy}{dx}$

M1

Grad normal = $-\frac{1}{2}$

use of $m_1 m_2 = -1$

M1

CSO

A1

$$y = -\frac{1}{2}(x-1)$$

OE

A1

4

[7]

Q2.

(a) (i) $y = (x^2 - 3)e^x$

$\frac{dy}{dx} = (x^2 - 3)e^x + 2xe^x$

product rule

M1

A1

2

(ii) $\frac{d^2y}{dx^2} = (x^2 - 3)e^x + 2xe^x + 2xe^x + 2e^x$

product rule from their $\frac{dy}{dx}$

M1

A1

2

(b) (i) $\frac{dy}{dx} = 0$

$\Rightarrow e^x(x^2 + 2x - 3) = 0$

$e^x f(x) = 0$ from $\frac{dy}{dx} = 0$

M1

$e^x(x + 3)(x - 1) = 0$

attempt at factorising or use of formula

m1

$\therefore x = -3, 1$

first correct solution

A1

second correct solution, and no others

SC No working shown:

$x = -3$ B2, $x = 1$ B2

A1

4

(ii) $x = -3$ $y'' = -4e^x$ max (–0.2)

Condone slip

M1

$x = 1$ $y'' = 4e^x$ min (10.9)

A1

2

[10]

Q3.

(a) $\frac{dy}{dx} = 3\sec^2 3x$

M1

for sec 3x

SC / 3sec² x *B1*

A1

Alternative

Use of product / Quotient rule (M1)

Good attempt

$\frac{3\cos^2 3x + 3\sin^2 3x}{\cos^2 3x}$ (A1)

Correct

2

(b) $\frac{dy}{dx} = \frac{(2x+1)3 - 2(3x+1)}{(2x+1)^2} = \frac{6x+3-6x-2}{(2x+1)^2}$

use of quotient rule

M1 A1

$= \frac{1}{(2x+1)^2}$

AG (no errors)

A1

Alternative

$-2(3x + 1)(2x + 1)^{-2} + 3(2x + 1)^{-1}$ (M1A1)

$= \frac{1}{(2x+1)^2}$ (A1)

$y = \frac{3}{2} - \frac{1}{2}(2x + 1)^{-1}$ *M1A1*

$\frac{dy}{dx} = (2x + 1)^{-2}$ *A1*

$= \frac{1}{(2x+1)^2}$

AG

3

[5]

Q4.

(a) $y = (3x-1)^{10}$

$\frac{dy}{dx} = 10(3x-1)^9 \times 3$

M1 for $a(3x-1)^9$ *where a = constant*

M1A1

2

$= 30\ (3x-1)^9$

(b) $\int x(2x+1)^8\,dx$

$u = 2x + 1$

$du = 2\,dx$

OE

B1

$\int = \int\left(\frac{u-1}{2}\right)u^8\left(\frac{du}{2}\right)$

all in terms of u. Condone omission of du

M1

$= \frac{1}{4}\int u^9 - u^8\,du$

$= \frac{1}{4}\left[\frac{u^{10}}{10} - \frac{u^9}{9}\right]$

$p\frac{u^{10}}{10} + q\frac{u^9}{9}$

B1

$= \dfrac{(2x+1)^{10}}{40} - \dfrac{(2x+1)^9}{36} (+c)$

OE; CAO

SC: correct answer, no working/parts in x (B1)

A1

4

[6]

Q5.

(a) $y = x \ln x$

$\dfrac{dy}{dx} = x \times \dfrac{1}{x} + \ln x$

use of product rule (only differentiating, 2 terms with + sign)

M1

$= \ln x + 1$

A1

2

(b) $\int (\ln x + 1) \, dx = x \ln x$

OE; attempt at parts with

M1

$\int \ln x \, dx = x \ln x - x (+c)$

A1

2

(c) $\int_1^5 \ln x \, dx = [x \ln x - x]_1^5$

$= (5 \ln 5 - 5) - (1 \ln 1 - 1)$

correct substitution of limits into their (b) provided $\ln x$ is involved

M1

$5 \ln 5 - 4$

ISW

A1

2

[6]

Q6.

(a) $x = \frac{1}{2}$ $y = \frac{\pi}{2}$ $\left(\text{or } 1.57, \sin^{-1} 1\right)$

ignore 90°

B1

1

(b) (i) $y = \sin^{-1} 2x$

$\sin y = 2x$ and $\frac{1}{2}\sin y = x$

AG (be convinced)

B1

1

(ii) $\frac{dx}{dy} = \frac{1}{2}\cos y$

B1

1

(c) $\frac{dy}{dx} = \frac{2}{\cos y}$

M1 for $\frac{k}{\cos y}$

M1A1

$\sin y = 2x$ and $\sin^2 + \cos^2 = 1$

use of to get $\cos y$ or $\cos^2 y$

M1

$\cos y = \sqrt{1 - 4x^2}$

$\frac{dy}{dx} = \frac{2}{\sqrt{1 - 4x^2}}$

AG; condone omission of proof of sign

A1

4

[7]

Q7.

(a) $x = 1$ $y^2 - y + 3 - 5 = 0$

M1

$(y-2)(y+1)=0$

Attempt to solve quadratic equation with $x = 1$

M1

$y = 2$ $y = -1$

A1

3

(b) (i) $2y\frac{dy}{dx} - x\frac{dy}{dx} - y + 6x = 0$

$+6x$; $-5 \rightarrow 0$

B1B1

Chain rule

B1

Product rule (M1 two terms)

M1A1

$6x - y + (2y - x)\frac{dy}{dx} = 0$

Factorise and obtain answer given

A1

6

Alternative

$\frac{dy}{dx}(y-x)^2 = (y-x)(0-6x)$

$5 \rightarrow 0$

(B1)

$-6x$

(B1)

$-(5-3x^2)\left(\frac{dy}{dx}-1\right)$

Recognisable attempt at quotient rule

(M1)

Completely correct OE

(A1)

$$\frac{dy}{dx}\left[(y+x)^2+(5-3x^2)\right]=(y-x)(-6x)$$
$$+(5-3x^2)$$

Factorise out $\frac{dy}{dx}$

(A1)

Given answer

Correct answer from correct working

Be convinced

(A1)

(ii) $(1, 2)$ $\frac{dy}{dx}=-\frac{4}{3}$

Substitute x = 1 and one y value from (a)

M1

$(1, -1)$ $\frac{dy}{dx}=\frac{7}{3}$

Both; follow on candidates y s

OE $\frac{-7}{-3}$ *; 3SF*

A1F

2

(iii) $y-6x=0$

B1

$(6x)^2-x\times 6x+3x^2-5=0$

M1

$36x^2-6x^2+3x^2-5=0$

$33x^2-5=0$

AG convincingly obtained

A1

3

[14]

Q8.

(a) $y = x \sin 2x$

$\frac{dy}{dx} = x\,2\cos 2x + \sin 2x$

M1

product rule

A1,A1

3

(b) (i) $y = (x^2 - 6)^4$

$\frac{dy}{dx} = 4(x^2 - 6)^3\,(2x)$ (or better)

M1 for $(x^2 - 6)^3$

M1A1

2

(ii) $\int 8x\,(x^2 - 6)^3\,dx = (x^2 - 6)^4$

for $c\,(x^2 - 6)^4$ if correct attempt

M1

$\int = \frac{1}{8}(x^2 - 6)^4\ (+c)$

A1

for $\frac{1}{k}(x^2 - 6)^4$ at 'by parts' M1A0

for $k = 8$

OR

$(x^2 - 6)^3 = x^6 - 18x^4 + 108x^2 - 216$ (M1A1)

$\int x\,(x^2 - 6)^3 = \frac{x^8}{8} - 3x^6 + 27x^4 - 108x^2$ (A1)

A1

3

[8]